Fiscal Year:	FY 2011	**Task Last Updated:**	FY 02/14/2011
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Levitation Observation of Dendrite Evolution in Steel Ternary Alloy Rapid Solidification (LODESTARS)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Web Page:	https://		
City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	04/03/2008	**End Date:**	04/02/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Received extension until 4/02/2012 per NSSC (Ed., 4/15/2011) NOTE: Grant may be extended, per P. Downey/MSFC (Ed., 2/18/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hyers, Robert (University of Massachusetts, Amherst)		
Grant/Contract No.:	NNX08AL21G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Undercooling in containerless processing has shown promise in producing metastable phases and materials that exhibit improved chemical homogeneity and ultra-fine grain sizes. Stainless steel alloys require special handling in order to achieve a significant undercooling range. In particular, chromium is particularly susceptible to contamination through reaction with crucible material. When attempting to perform rapid solidification tests, the formation of active heterogeneous nucleation sites limits the undercooling that may be achieved. By using a containerless technique, this contamination is minimized. The focus of this program is to study of the effect of microgravity on metastable solidification. By observing changes in the mechanism for nucleation of the stable phase following primary recalescence from undercooled melts, the role of convection in phase selection may be evaluated. The value of this research is that the results have application to the design of industrial welding, spray forming and strip casting operations for a commercially important class of structural materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The value of this research is that the results have application to the design of industrial welding, spray forming and strip casting operations for a commercially important class of structural materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Task Progress:	Highlights of Work Accomplished During Grant Period 2010-11. • ESA proposals passed science peer review and are approved 6/1/2010 o PARSEC - Peritectic Alloy Rapid Solidification with EM Convection o ThermoLAB - Thermophysical Properties of Liquid Metallic Alloys • Invited presentation at 1st meeting of the ESA International Topical Team on Solidification (SOL-EML) in Cologne DE 7/8/10 • Hired Tufts PhD student Rongbiao Gu to conduct flight support testing at MSFC • Hired Tufts MS student Paul Sander to conduct ESL tests at MSFC • Briana Tomboulian MS/PhD student at UMass to conduct flow modeling tasks • Initiated sample preparation procedures for ground-based EML testing of evaporation rates and coupling coefficients for Fe-Cr-Ni ThermoLAB samples • Delivered 5 simulated flight ThermoLAB samples to DLR MUSC 8/2010 • Delivered samples for MSFC/ELS thermophysical property evaluation 7/15/2010 • Conducted ESL baseline tests on Ti-Al-Nb alloys at MSFC 8/9-13/2010 • Attended 3rd MSL-EML Working Group at ESA/ESTEC 11/15/2010 • Invited member ESA International Topical Team on Thermophysical Properties (ThermoLab-ISS kick-off meeting) at ESA/ESTEC 11/21/2010 • Conducted ESL viscosity and density tests on FeCo alloys at MSFC 1/10-16/2011 • Obtained approval from Wiley-VCR as editor of a solidification text involving collaboration with 22 international colleagues from our ESA topical teams.
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Abstracts for Journals and Proceedings	Hyers RW. "Computer-aided Experiments in Containerless Processing of Materials." Reactive Metals Workshop, Seattle, WA, February 18, 2010. Reactive Metals Workshop, Seattle, WA, February 18, 2010. , Feb-2010
Abstracts for Journals and Proceedings	Hyers RW. "Computer-aided Experiments in Containerless Processing of Materials." Invited presentation, Saint-Gobain, Northboro, MA, February 25, 2010. Saint-Gobain, Northboro, MA, February 25, 2010. , Feb-2010
Abstracts for Journals and Proceedings	Hyers RW. "Modeling and Computer-aided Experiments in Non-contact Materials Processing." Invited presentation, International Topical Team: Solidification in Electromagnetic Levitation (SOL-EML), Deutsches Zentrum für Luft- und Raumfahrt Köln, Cologne, Germany, July 9, 2010. International Topical Team: Solidification in Electromagnetic Levitation (SOL-EML), Deutsches Zentrum für Luft- und Raumfahrt Köln, Cologne, Germany, July 9, 2010. , Jul-2010
Abstracts for Journals and Proceedings	Matson DM. "Nucleation Within the Mushy Zone." Invited presentation, International Topical Team: Solidification in Electromagnetic Levitation (SOL-EML) (SOL-EML), Deutsches Zentrum für Luft- und Raumfahrt Köln, Cologne, Germany, July 9, 2010. International Topical Team: Solidification in Electromagnetic Levitation (SOL-EML) (SOL-EML), Deutsches Zentrum für Luft- und Raumfahrt Köln, Cologne, Germany, July 9, 2010. , Jul-2010
Articles in Peer-reviewed Journals	Matson DM, Hyers RW, Volkmann T. "Peritectic alloy rapid solidification with electromagnetic convection." Journal of the Japanese Society of Microgravity Applications, 2010;27(4). Invited paper. , Dec-2010
Papers from Meeting Proceedings	Matson DM. "Status of the LODESTARS and ELFSTONE Programs." Invited presentation, THERMOLAB Workshop, ESA ESTEC, Noordwijk, Netherlands, November 19, 2010. THERMOLAB Workshop, ESA ESTEC, Noordwijk, Netherlands, November 19, 2010. , Nov-2010